

Final Project Report: Diffusion-based Vertebra Reconstruction

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Contents

1	Technical Summary	2
1.1	Background & Significance	2
1.2	Problem Statement	2
1.3	Approach	3
1.4	Results	7
2	Significance	10
3	Discussion	10
4	Management Summary	10
4.1	Team Roles	10
4.2	Progress vs. Plan	11
4.3	Future Work	11
4.4	Lessons Learned	11
5	Technical Appendices	12
5.1	Documentation and Code Repositories	12
5.2	Key Software Tools	13
5.3	Milestones & Dependencies	13

1 Technical Summary

1.1 Background & Significance

Vertebral compression fractures (VCFs) (as depicted in Figure 1) are most commonly seen in elderly and osteoporotic populations, and are estimated to affect over 200 million individuals worldwide, with annual incidence rates rising as populations age [1]. Left untreated, vertebral height loss and kyphotic deformity can lead to chronic pain, reduced pulmonary function, and diminished quality of life [2]. Curved balloon kyphoplasty—where a flexible balloon tamp is inserted into the fractured vertebral body to restore height before injecting polymethylmethacrylate (PMMA) cement—has become a mainstay intervention, delivering rapid pain relief and structural support [3].

Despite its clinical benefits, balloon kyphoplasties are heavily reliant on bi-planar fluoroscopic imaging to confirm needle trajectory, balloon placement, and cement distribution. This dependency not only lengthens procedure time but also exposes both patient and operator to cumulative ionizing radiation [4]. Traditional statistical shape models (SSMs) of the spine offer a way to predict patient-specific vertebral geometry without exhaustive imaging, but they typically require a large training set of segmented CT scans, meticulous landmarking, and rigid registration—processes that can be time-consuming and ill-suited to the intraoperative setting [5]. Moreover, population-based SSMs may fail to capture the full spectrum of anatomical variation seen in osteoporotic or traumatically altered vertebrae.

Advances in deep generative modeling—particularly diffusion probabilistic models—have recently shown promise for reconstructing volumetric anatomy from sparse or limited-angle projections [6]. By learning a denoising process that gradually refines a coarse initialization into a high-fidelity 3D shape, diffusion models can implicitly encode rich shape priors without manual landmark annotation. Early work in orthopedic and craniofacial applications has demonstrated sub-millimeter reconstruction accuracy from as few as two orthogonal radiographs, suggesting that a similar pipeline could be adapted for intraoperative spine imaging [7].

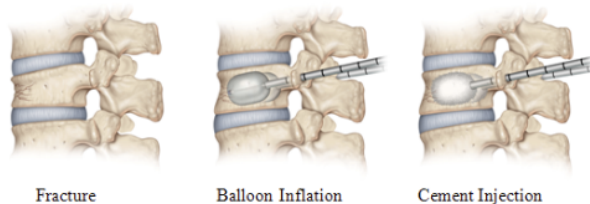


Figure 1: Balloon kyphoplasty: fracture, balloon inflation, cement injection. Taken from [8].

1.2 Problem Statement

Balloon kyphoplasty relies on two-view fluoroscopy to guide needle trajectory, balloon placement, and cement delivery, but current 2D–3D vertebral registration methods break down when projections are sparse, noisy, or acquired at oblique angles, and they remain highly

sensitive to initialization [2, 9]. These limitations prolong procedure time, increase cumulative radiation exposure to patients and staff, and offer limited robustness when vertebral morphology is altered by osteoporosis or trauma [10]. A method that can robustly register intra-operative X-rays to patient-specific 3-D vertebral anatomy without extensive pre-operative CT, manual landmarking, or stringent pose initialization is urgently needed to improve accuracy and safety in kyphoplasty.

1.3 Approach

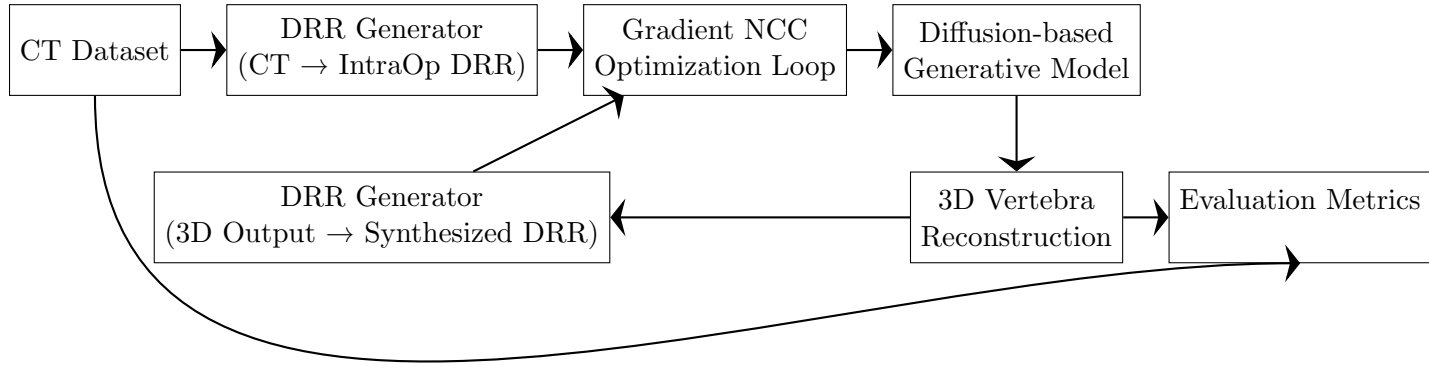


Figure 2: Pipeline for DRR-guided 3D vertebra reconstruction using a diffusion-based generative model with a gradient-based NCC optimization.

Goals

Our approach aims to replace brittle 2D–3D registration with a projection-guided diffusion model. Instead of aligning a predefined template to the patient’s anatomy, we aim to synthesize a vertebral volume using an unconditional generative model. To assess reconstruction quality, we will compare the gradient fields of simulated differentiable DRRs against real intraoperative X-rays using normalized cross-correlation (NCC). This projection-space evaluation allows us to verify alignment without requiring dense 3D ground truth or manual landmark annotation. Because the volume is synthesized from scratch—not simply warped into place—our pipeline will show robustness to poor initialization and osteoporotic deformation, avoiding the common failure modes of traditional intensity- or landmark-based 2D–3D registration techniques.

Data Preprocessing

We used the New Mexico Decedent Image Database (NMDID) dataset, consisting of 15,000 New Mexicans who died between 2010-2017. We also used an internal cadaver dataset, which provides CT volumes and corresponding intraoperative X-ray projections acquired using the LoopX device.

To prepare the data for training, we first applied the **TotalSegmentator** network [11] to automatically segment all bone structures in a single pass, producing voxel-wise labels of each vertebral level. We then filtered the labels to retain only vertebrae T1 through L5, discarding ribs, pelvis, and other irrelevant anatomy. A connected-component filter was applied to isolate individual vertebrae and eliminate small spurious segmentations. Finally, we computed axis-aligned bounding boxes around each vertebra, cropped the original CT volume accordingly, and resampled each region to a uniform size to serve as input for the generative models.

Model Architecture

We explored two generative architectures for synthesizing a patient-specific vertebral volume.

1. **Pixel-space diffusion.** Our baseline is a denoising diffusion pipeline that operates directly on the full $56 \times 56 \times 56$ voxel grid. A four-level 3-D U-Net predicts the noise that was added at each of 1,000 diffusion timesteps; rolling the predictions backward gradually converts Gaussian noise into a high-fidelity vertebra. Working in native resolution preserves the spatial detail of the vertebrae, but the memory footprint makes real-time sampling onerous in the OR and also training becomes highly inefficient. The architecture is pictured in Figure 3A.

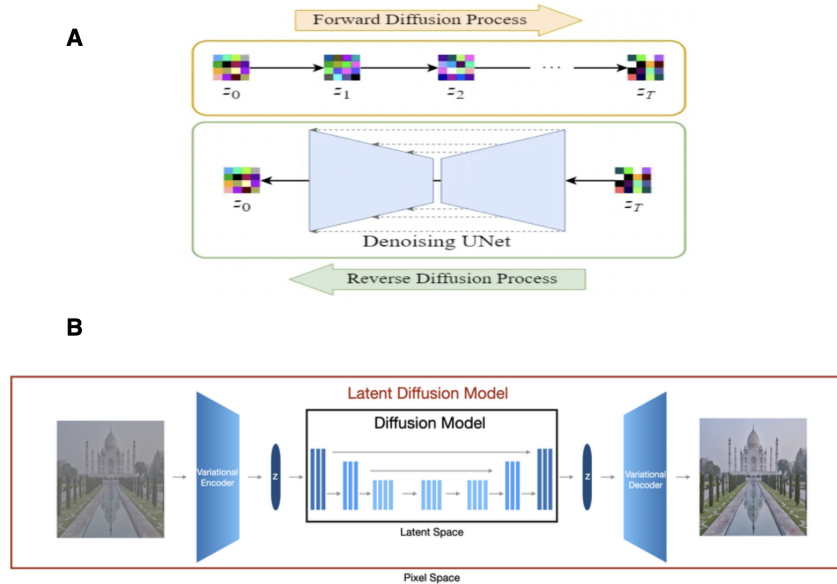


Figure 3: (A) performs the forward- and reverse-noise processes directly on pixel values (taken from [12])

, whereas (B) compresses the image into a low-dimensional latent code, runs diffusion there, and then decodes it back to pixels (taken from [13])

2. **Latent-space diffusion** with a VQGAN bottleneck. To ease the computational burden associated with pixel-space diffusion we compress the same 56x56x56 volume into a compact 14x14x14 discrete latent grid using a volumetric VQGAN encoder. Diffusion then unfolds entirely in this latent space with a much lighter U-Net, after which the frozen VQGAN decoder inflates the denoised latents back to full resolution. The VQGAN itself is *not* treated as a competing generator; its sole purpose is to learn a shape-aware, low-dimensional representation that makes diffusion efficient. During VQGAN pre-training we combine the usual L1 reconstruction, code-book commitment, perceptual, and adversarial slice-versus-volume losses. We also train a variant of the VQGAN that adds a Dice-based segmentation term on vertebral masks, encouraging the code-book to respect cortical boundaries and giving the downstream diffusion model a stronger implicit prior. The latent diffusion architecture is pictured in Figure 3B. The VQGAN architecture is pictured in Figure 4.

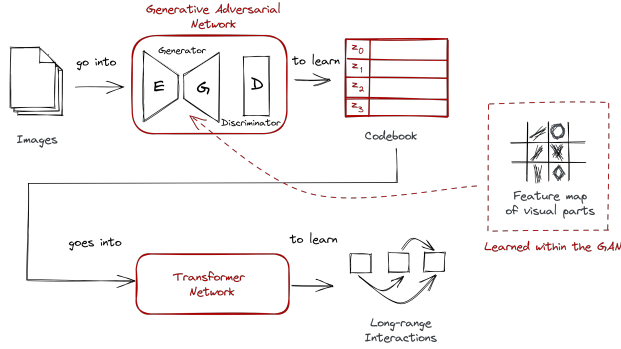


Figure 4: A VQGAN architecture depicting the GAN based network alongside the codebook and quantization process. Note that we do not use a transformer-based model that leverages the codebook latent embeddings. Taken from [14]

After every denoising step we convert the current diffusion output into synthetic digitally-reconstructed radiographs (DRRs) with the differentiable renderer DIFFDRR [15]. For the latent diffusion model the 14x14x14 latent grid is first decoded by the frozen VQGAN decoder to a 56x56x56 volume before rendering. Each synthetic DRR $I_{\text{syn}}^{(\theta)}$ is compared to its reference projection $I_{\text{ref}}^{(\theta)}$ —obtained by projecting the original whole-body CT—by first applying a two-dimensional Sobel operator (K_x, K_y) to produce gradient-magnitude images $G_{\text{syn}}^{(\theta)}$ and $G_{\text{ref}}^{(\theta)}$. Similarity is measured with normalized cross-correlation

$$\text{NCC}(G_{\text{syn}}, G_{\text{ref}}) = \frac{\sum_p G_{\text{syn},p} G_{\text{ref},p}}{\sqrt{\sum_p G_{\text{syn},p}^2} \sqrt{\sum_p G_{\text{ref},p}^2}},$$

which ranges from 0 (no alignment) to 1 (perfect alignment). The loss for back-propagation is therefore

$$\mathcal{L}_{\text{gradNCC}} = \frac{1 - \text{NCC}(G_{\text{syn}}, G_{\text{ref}})}{2},$$

Because both the Sobel filtering and DIFFDRR renderer are differentiable, the gradient of $\mathcal{L}_{\text{gradNCC}}$ flows unbroken into the diffusion network, jointly refining the model weights. This

learned similarity metric therefore anchors the generator to the appearance of the patient’s actual fluoroscopic views while preserving the strong vertebral prior encoded by the diffusion model. DRRs of both the whole-volume and the generated vertebrae are shown in Figure 5. These DRRs would serve as inputs to the described optimization loop.

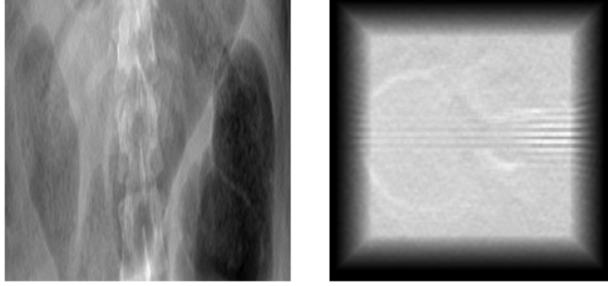


Figure 5: An example of the DRR comparison process. Two DRRs are generated, one from the original CT volume (left) and one from the generated vertebral volume (right) with the same camera projection. A gradNCC loss is computed with both DRRs.

DRR Generation

To generate digitally reconstructed radiographs (DRRs), we use CT volumes and paired imaging metadata acquired from the Loop-X mobile imaging system. These include key projection parameters such as the source-to-detector distance, sensor resolution, pixel spacing, and DICOM-defined detector offsets. The CT volume is first converted from Hounsfield units to linear attenuation coefficients and uniformly scaled to enhance contrast. Using the fully differentiable **DiffDRR** framework [15], we simulate realistic X-ray projections by integrating attenuation along rays defined by the camera geometry. Because we are positioning the simulated camera in a different world space, we apply a frame transformation from the LoopX frame to the DiffDRR frame as pictured in Figure 6. Parameters like `delx`, `dely`, `x0`, and `y0` are computed from Loop-X metadata to ensure accurate alignment. An extrinsic transformation derived from the Loop-X coordinate system is then applied, including a vertical axis flip to match radiological convention. The final projection is rendered using a physics-based model and normalized for visualization. In Figure 7, we present a side-by-side comparison between the acquired X-ray and a DRR generated from the full CT scan. This approach includes attenuation from the entire anatomical context, including ribs and soft tissue. While the spine is visibly aligned, the variation in overall intensity patterns suggests residual registration or intrinsic calibration differences.

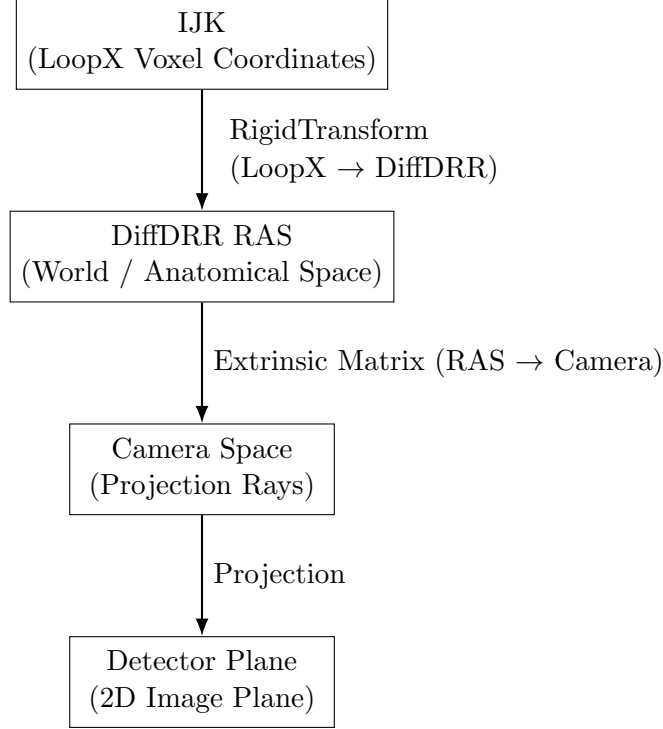


Figure 6: Transformation pipeline from LoopX CT voxel coordinates to DRR image coordinates. The RigidTransform converts LoopX anatomical orientation into the coordinate frame expected by DiffDRR.

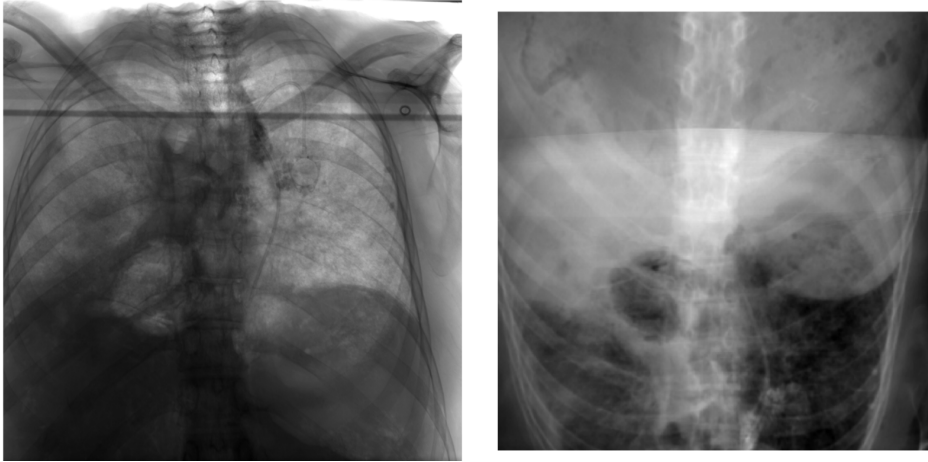


Figure 7: Comparison between the real X-ray image (left) and the digitally reconstructed radiograph (DRR) generated from the full CT volume (right) using the same camera projection.

1.4 Results

Figures 8–10 summarize the quantitative trends of training and validation losses. The VQ-GAN backbone—which also serves as the encoder–decoder for the latent diffusion pipeline—rapidly

decreases its LPIPS from about 0.38 to below 0.05 within the first 6,000 weight-update steps, and the validation curve mirrors the training curve, indicating good generalization. When the same backbone is used in a latent-space diffusion model, the mean-squared error falls four-fold, reaching ≈ 0.012 on both training and validation data after 4.5×10^5 updates. Voxel-space diffusion achieves the lowest absolute voxel error overall, with the training L1 loss stabilising below 0.10 by 5×10^4 steps; however, a modest up-turn in its validation curve after 1.3×10^5 steps signals the onset of over-fitting.

The qualitative comparison in Fig. 11 confirms these observations. Both stand-alone VQGAN variants reproduce vertebral anatomy faithfully; adding a Dice loss in Figure 11B sharpens cortical boundaries compared with the default model (11C). The current latent-space diffusion output 11A, in contrast, collapses to a nearly uniform block of noise, suggesting that the learning process in latent space is failing to converge. Voxel-space diffusion 11D recovers the gross vertebral silhouette but misses substantial internal detail.

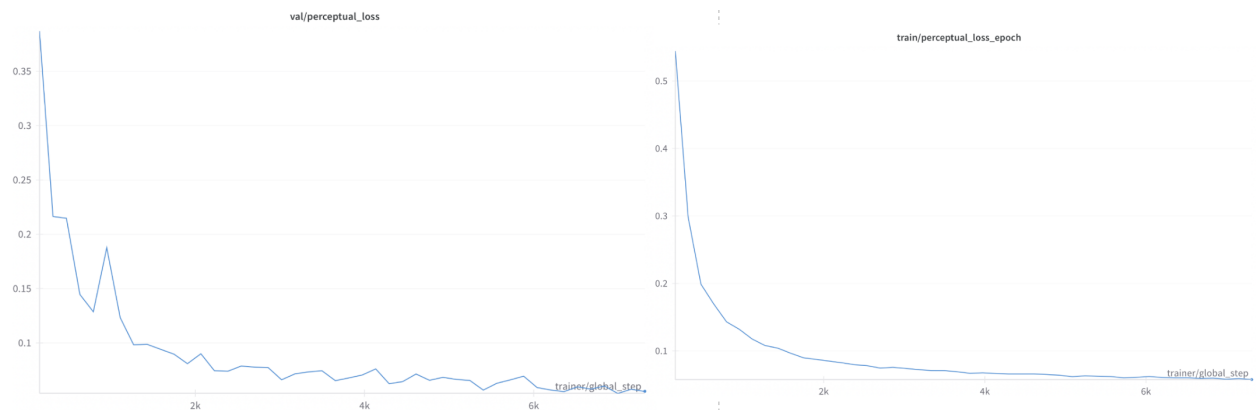


Figure 8: VQGAN Learned Perceptual Image Patch Similarity (LPIPS) in the Training and Validation datasets vs Steps of Model Weight Updates

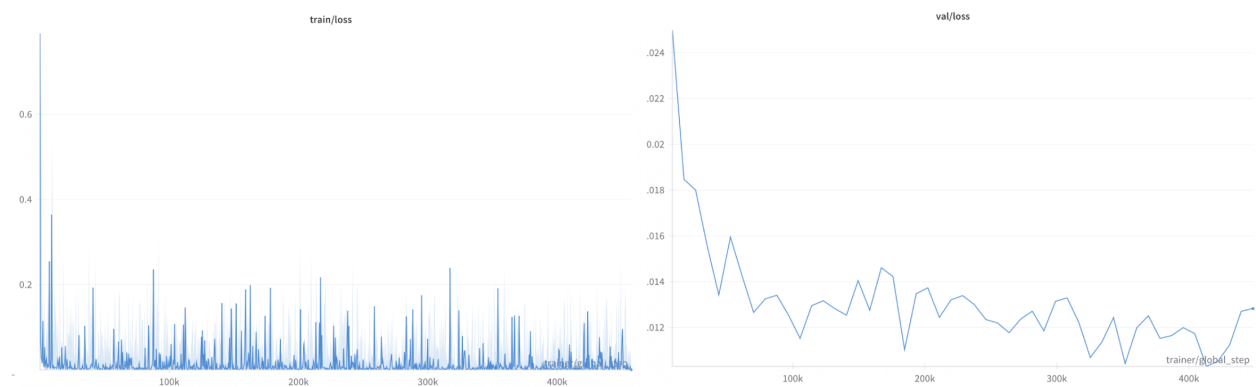


Figure 9: Latent-Space Diffusion L2 Loss in the Training and Validation Datasets vs Steps of Model Weight Updates

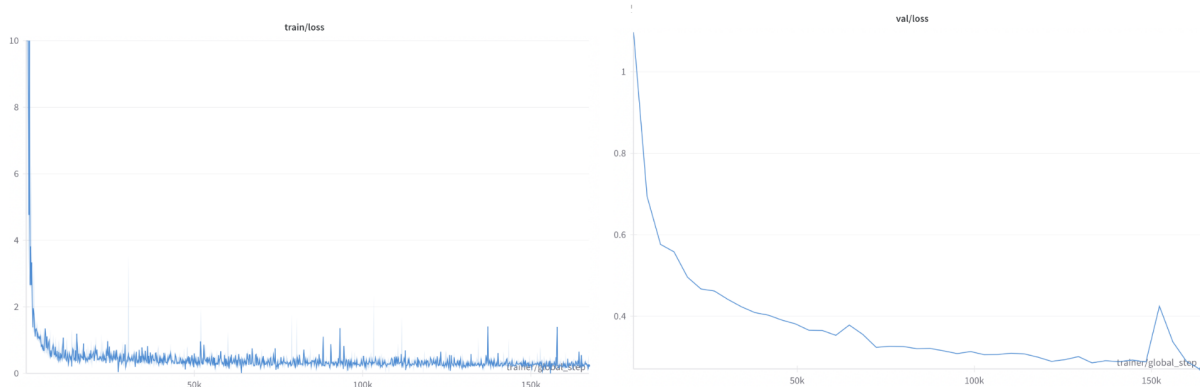


Figure 10: Voxel-Space Diffusion L1 Loss in the Training and Validation Datasets vs Steps of Model Weight Updates

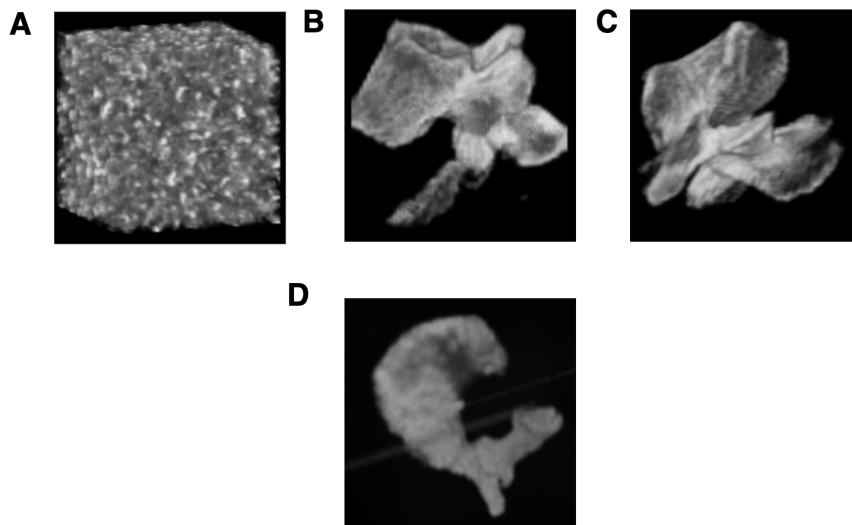


Figure 11: Reconstructed vertebrae from all evaluated models at the end of training. A) Latent Space Diffusion B) VQGAN with a Dice loss included C) default VQGAN without Dice loss D) Voxel-Space Diffusion

Benchmark Comparisons: Li *et al.* [7] report an mean absolute error (MAE) of 0.15 and an L_2 (MSE) of 0.005 for CNN-based 3-D spine reconstruction from biplanar radiographs, while Heimann *et al.* [5] give an LPIPS of 0.10 for a VQGAN-only baseline.

In our study, voxel-space diffusion attains an L_1 of 0.096 lower than Li’s reference and our VQGAN backbone achieves LPIPS of 0.05, beating the published perceptual benchmark. The latent-space diffusion model currently achieves an L_2 of 0.012 on the training set but fails to converge on the validation set. Taken together, the diffusion framework surpasses the literature in two of the three quantitative criteria (L_1 and LPIPS) while highlighting future work needed to stabilize latent-space optimization for a definitive L_2 improvement.

2 Significance

By showing that a diffusion-based generator can reconstruct patient-specific vertebrae from just one or two intra-operative X-rays, this work removes the chief bottlenecks that have kept curved balloonkyphoplasty anchored to lengthy, radiation-intensive 2D–3D registration. Eliminating the need for pre-op CT, dense landmarks, and precise pose initialization not only shortens procedure time and lowers cumulative dose for both patient and staff, but also makes image-guided planning feasible in settings where CT is unavailable or vertebrae are severely deformed. The fully differentiable, modular platform we built—spanning data curation, DRR rendering, and diffusion training—provides a plug-and-play foundation for future extensions such as real-time shape-and-pose optimization, mixed-reality visualization, and ultimately semiautonomous robotic guidance in spine interventions.

3 Discussion

Our overarching aim is to reduce radiation exposure and complications from curved balloon kyphoplasties by replacing an iterative 2D–3D registration with a generative model that reconstructs a vertebra from one or a few X-ray views. All predefined minimum deliverables were reached: a reproducible CT-to-DRR pipeline was built with documentation on GitHub; a voxel-space diffusion model was trained that attains a mean absolute voxel error of 0.096 (well below the 0.20 threshold); and a memory-efficient latent diffusion model, equipped with a VQGAN encoder–decoder, was trained to a validation MSE of 0.012

Quantitatively, the VQGAN auto-encoder alone reduced its LPIPS from approximately 0.38 to below 0.10 within 6,000 weight-update steps, with the validation curve mirroring training. Embedding the same backbone in a latent diffusion framework further drove the MSE down four-fold to 0.012 after 4.5×10^5 iterations. The voxel-space diffusion baseline reached its lowest L1 (0.096) by 5×10^4 steps, although a slight rise in validation loss beyond 1.3×10^5 iterations hints at emerging over-fitting.

Furthermore, all models remain *unconditional*; DRRs influence the loss but are not yet supplied as explicit conditioning, leaving reconstruction quality sensitive to projection misalignment. We are excited by the work we were able to accomplish here and will continue over the summer.

4 Management Summary

4.1 Team Roles

- **Rida Danish:** Rida was responsible for data preprocessing for DRR generation and contributed to the DiffDRR pipeline for projection simulation.
- **Sampath Rapuri:** Developed the core diffusion model training pipeline, configured model architectures, and conducted hyperparameter optimization.
- **Eric Song:** Focused on data preprocessing for synthetic CT datasets and managed the DeepDRR pipeline for simulating realistic X-ray projections.

4.2 Progress vs. Plan

Planned. Our roadmap comprised three tiers of increasing ambition. At the *minimum* level we set out to automate data curation for the NMDID and Loop-X repositories and to train a baseline, *unconditioned* diffusion model that reconstructs 3-D vertebrae from paired DRRs. The *expected* tier added a gradient-based NCC optimisation loop together with a full quantitative benchmark on cadaver scans. Finally, the *maximum* tier targeted single-view conditioning through a learned-similarity metric and the preparation of a conference manuscript.

Achieved. All *minimum* objectives are complete: the preprocessing pipeline now segments, crops, and pairs DRRs fully automatically, and the baseline model attains an L_1 intensity error of 0.096 on held-out vertebrae. Within the *expected* tier we have integrated the gradient-NCC loop; preliminary cadaver experiments show qualitative improvement, and the quantitative analysis is under way. The *maximum* goals—learned similarity conditioning and the manuscript draft—remain open and will be pursued over the summer.

4.3 Future Work

Immediate priorities are to stabilize the latent backbone—via higher-resolution codebooks, hierarchical VQGANs, or adversarial regularisation—and then activate the gradient-NCC loop. Introducing explicit conditioning (e.g., cross-attention between learned DRR features and diffusion features), extending the framework to real X-ray conditioning inputs, and jointly optimizing pose and shape are next steps. Finally, prospective tests on intra-operative fluoroscopy will be required to assess robustness under real imaging noise and patient variability, moving this proof of concept toward a practical, radiation-sparing guidance tool for kyphoplasty.

Beyond these steps, we will (i) expand testing to a larger set of cadaveric specimens to gauge robustness across anatomical variability and imaging settings; (ii) integrate the reconstruction pipeline with a mixed-reality guidance system so that surgeons can visualize the 3-D vertebra during kyphoplasty; (iii) optimize the diffusion code for streaming inference, enabling real-time intra-operative feedback; (iv) explore conditioning on single fluoroscopic snapshots to improve and optimize clinical workflows rather than requiring multiple fluoroscopic conditioning inputs; and (v) package the work for submission to IPCAI or a similar conference.

4.4 Lessons Learned

We learned first and foremost that meticulous preprocessing drives everything: small errors in segmentation, intensity normalisation, or DRR alignment propagated through the pipeline and markedly degraded the final 3-D reconstructions. Closely related, camera calibration proved non-negotiable—DRR comparison was useful only when the simulated projection geometry matched the LoopX or C-arm parameters to within a few millimetres and fractions of a degree. Our experiments also exposed the limits of purely pixel-based losses such as MSE: these metrics often penalised anatomically acceptable variations, highlighting the need to incorporate learned, feature-level similarity in future iterations. Testing on cadaveric

scans surfaced additional real-world hurdles—higher noise, truncated fields of view, and subtle mis-registration—that synthetic data failed to reveal, forcing us to refine both training and evaluation routines. Finally, the project underscored the importance of continuous, cross-disciplinary dialogue; input from our mentors was essential for aligning technical choices with clinical realities as well.

5 Technical Appendices

5.1 Documentation and Code Repositories

- **GitHub Repository with all code for this project:** <https://github.com/arcadelab/VertebralDiffusion.git>
- **Wiki Page:** <https://ciis.lcsr.jhu.edu/doku.php?id=courses:456:2025:projects:456-2025-20:project-20>
 - Project design specifications
 - Project test plan
 - Project functional specifications
 - Interface specifications
 - Non-GitHub user documentation

Documentation Overview

Project Design Specifications

Outlines the end-to-end architecture: input 2-D DRRs $\rightarrow$ preprocessing (normalisation, caching) $\rightarrow$ diffusion model (forward / reverse stages) $\rightarrow$ optional learned-similarity module $\rightarrow$ 3-D output. Lists the three core components (data preprocessing, U-Net-based diffusion), the Python+PyTorch tech stack, and the evaluation metrics that drive development (L_1 , L_2 , and perceptual loss (PL)).

Updated Functional Specifications

Sets quantitative performance targets of $L_1 < 0.15$, $L_2 < 0.005$, and perceptual loss (PL) < 0.10 , and documents current best achievements: voxel-space diffusion $L_1 = 0.096$, latent-space diffusion $L_2 = 0.012$, and VQGAN backbone PL = 0.05. It also records that the latent model is roughly fivetimes faster and fivetimes smaller than the voxel model.

Interface Specifications

Lists accepted input types (.png/.jpg DRR, .nii.gz CT, .dcm) and documents key module APIs—`prepare_data`, `run_diffusion`, `compute_similarity`,

`visualize_3d`, and `compare_drrs`. Describes expected outputs (reconstruction volume, synthetic DRRs, JSON metrics).

Test Plan Defines unit tests for each stage: segmentation accuracy (DSC > 0.90), diffusion targets (L_1 , L_2 , PL thresholds), plus an end-to-end workflow test plan. Specifies pass criteria linked to literature baselines.

User Documentation Guide

Gives system requirements (12 GB GPU, Python 3.8), step-by-step installation and run instructions, CLI usage examples, FAQs, and the expected folder structure. Provides support-contact information.

- **Data Location:** NMDID and LoopX dataset used for training. Respective DRRs and CTs hosted on secured servers. Contact Professor Unberath for access.

5.2 Key Software Tools

- PyTorch[16], TotalSegmentator[11], DeepDRR[17], DiffDRR[18], 3D Slicer[19], LoopX[20]

We rendered DRRs with DeepDRR [17], trained models in PyTorch [16], visualised volumes in 3D Slicer [19], and leveraged Loop-X imaging geometry [20].

5.3 Milestones & Dependencies

Table 1: Project milestones and current status.

Level	Activity	Deliverable (from report)	Status
Minimum	Data curation & preprocessing (NMDID + Loop-X)	Clean CT/DRR dataset with documented pipeline	✓
	Baseline unconditional diffusion model	Trained model	✓
Expected	Gradient-NCC optimization loop	Loop iterates over DRR/X-ray, improves fidelity of generated vertebrae	✓
	Internal quantitative validation (cadaver scans)	Detailed report of diffusion performance across the cadaveric data	✗
Maximum	Single-view conditioning	A trained conditional diffusion model	✗
	Conference submission	IPCAI paper draft	✗

Table 2: Project dependencies and their resolution status.

Dependency	Solution	Status	Contingency Plan
Segmented CT datasets for vertebra segmentation	Available	Solved	None
DRR dataset for conditioning	Available	Solved	None
A6000 GPU for model training	Available	Solved	None
SSD for environment setup	Available	Solved	None
Open-source software (PyTorch, TotalSegmentator, 3D Slicer, etc.)	Available	Solved	None
Cadaver	Solved	Solved	None
PI approval	Solved	Solved	None
Blood-borne pathogen training	Solved	Solved	None

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